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PIP_Decomp Appendix: Testing + Validation

1. COMPARING PIP_DECOMP DECAY FITS TO LOG-LINEAR FITTING

The following is a summary of the tests performed with artificial data to investigate the constraints and vulnerabilities of PIP_Decomp's assumption of repeated gaussian injections. A common means of estimating decay times in observed light curves is to fit an appropriate range of the decaying curve with a log-linear regression, typically from the time that the flux (or some smoothed or averaged flux curve) is at 90% of its peak value to the time that it falls to 10%. To compare the application's performance against this common log-linear fitting approach, as well as to verify its ability to reproduce input parameters from a known model, sets of simulated light curves of one or two injections have been created as follows:

- A 1-injection test with varying ratio of impulse to decay timescale to examine the quality of the fits and the effect of noise on the fitting parameters.
- A test of 2-injection fitting with noise, in which the first injection's decay time is varied to evaluate the effects that the decay of the first injection has on the measurement of the decay time of the second injection.

Each will be fit with PIP_Decomp, and with a log-linear fit of the decays between peaks and troughs marked by finding local extrema in a 4-time step rolling average of the data and fitting the central 80% of the flux difference between the maxima and the following minima.

1.1. *One injection with changing τ*

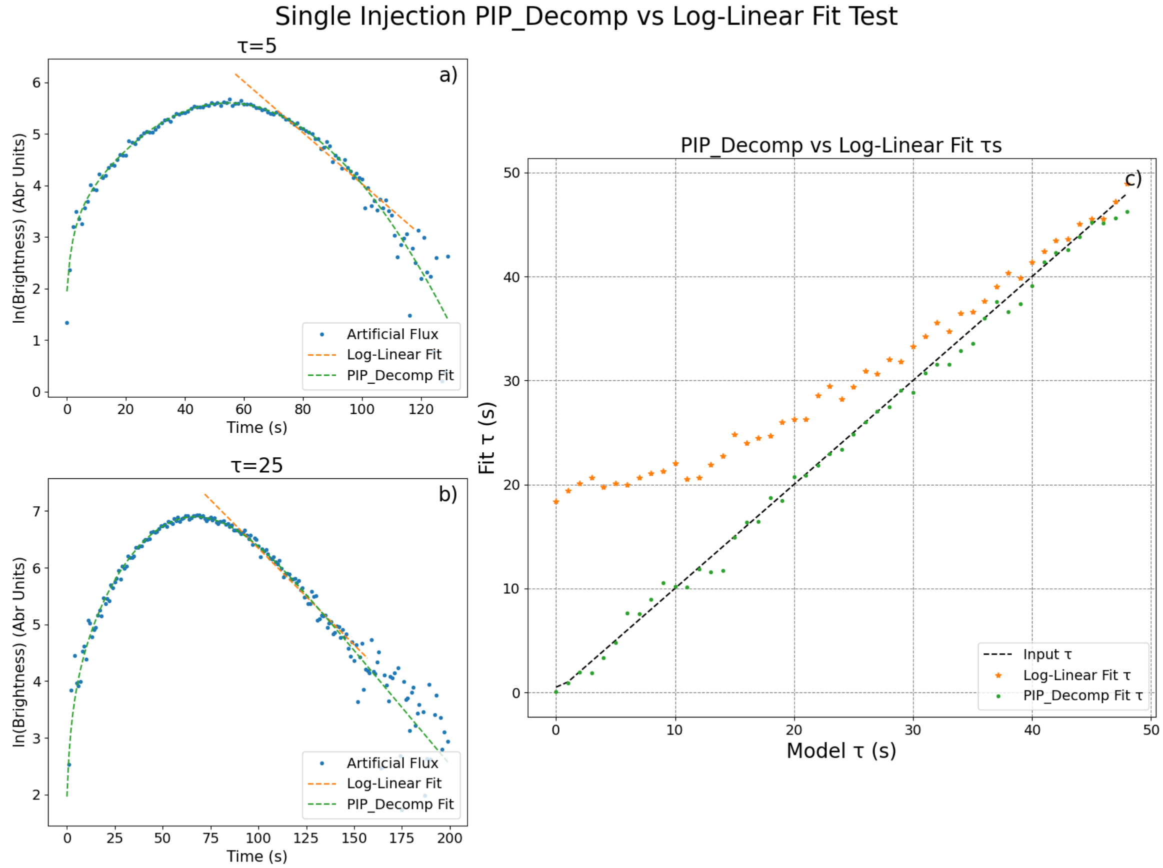

Figure 1. Results of fitting simulated 1-injection light curves ($a = 50$ SFU/s, $\mu = 50$ s, $\sigma = 25$ s, $\tau = k$ s with $k = [0, 49]$), with added noise, vs log-linear fitting. a) An example light curve on a log-linear scale at τ=5 s, short compared to the injection time. The simulated data (points) are overlaid with the PIP_Decomp fit (dashed green line) and the log-linear estimate (dashed orange line). b) Same as a, for a lightcurve at τ=25 s, the case of equal injection and decay times. c) The fitted decay tau_1 for PIP_Decomp (dots) and for the log-linear regression method (star symbols) as a function of varying τ_1. The line shows the decay parameter used in the simulation.

Following a successful sanity test of noiseless data to ensure that PIP_Decomp model functions exactly reproduce the input parameters, a further test of PIP_Decomp performance was undertaken. For this test, Gaussian noise of ±5% of the maximum amplitude of a given curve was added to simulated light curves built from PIP_Decomp's 1-injection model. The 5% noise level is much higher than the signal-to-noise ratio on a typical flare (from EOVSA, for instance), but it is done for a clearer effect. For each 1-injection test, the injection parameters are kept constant at $a = 50$ SFU/s, $\mu = 50$ s, $\sigma = 25$ s, but the decay time is varied according to $\tau = k$ s with $k = [0, 49]$ to cover a range of ratios of injection time σ and decay time τ.

Two example light curves from the sequence are shown in Figure 1, one with a short decay time $tau \ll \sigma$ and one where the decay and injection times are equal. Each curve is fit by both PIP_Decomp (green dashed curve) and a simple log-linear regression between the 90% and 10% flux levels (orange dashed line). Figure 1c compares the simulated decay time (black dashed line) with the parameters from PIP_Decomp (green points) and the log-linear regression (orange points) for the entire sequence of decay times. As expected, the log-linear approach returns a "decay" time close to the injection time for small decay-to-injection-time ratios and only asymptotically approaches the correct decay time at large decay-to-injection-time ratios, while PIP_Decomp faithfully measures the true decay time over the entire sequence.

1.2. *Two injections with changing* τ_1

For the 2-injection test, a constant second injection and decay with parameters $a_2 = 15$ SFU/s, $\mu_2 = 80$ s, $\sigma_2 = 10$ s, $\tau_2 = 25$ s is added to the same changing flux as for the one injection test, parameters $a_1 = 12$ SFU/s, $\mu_1 = 20$ s, $\sigma_1 = 15$ s, and $\tau = k$ s with $k = [0, 49]$.

In Figure 2 the behavior of the 1-injection test is still seen for the varying first injection, with the poor showing for the linear regression method at short decay times. But the addition of the static second injection's flux toward the end of the first even more significantly increases the log-linear estimation of τ_1 at larger τ_1 (orange points in Figure 2c), as it both raises and delays the minimum used, increasing the apparent decay time. The extent to which this occurs will depend on the time separation of the two injections relative to the injection widths and decay times. The second injection's fit shows an additional weakness of the linear method. Where the decay

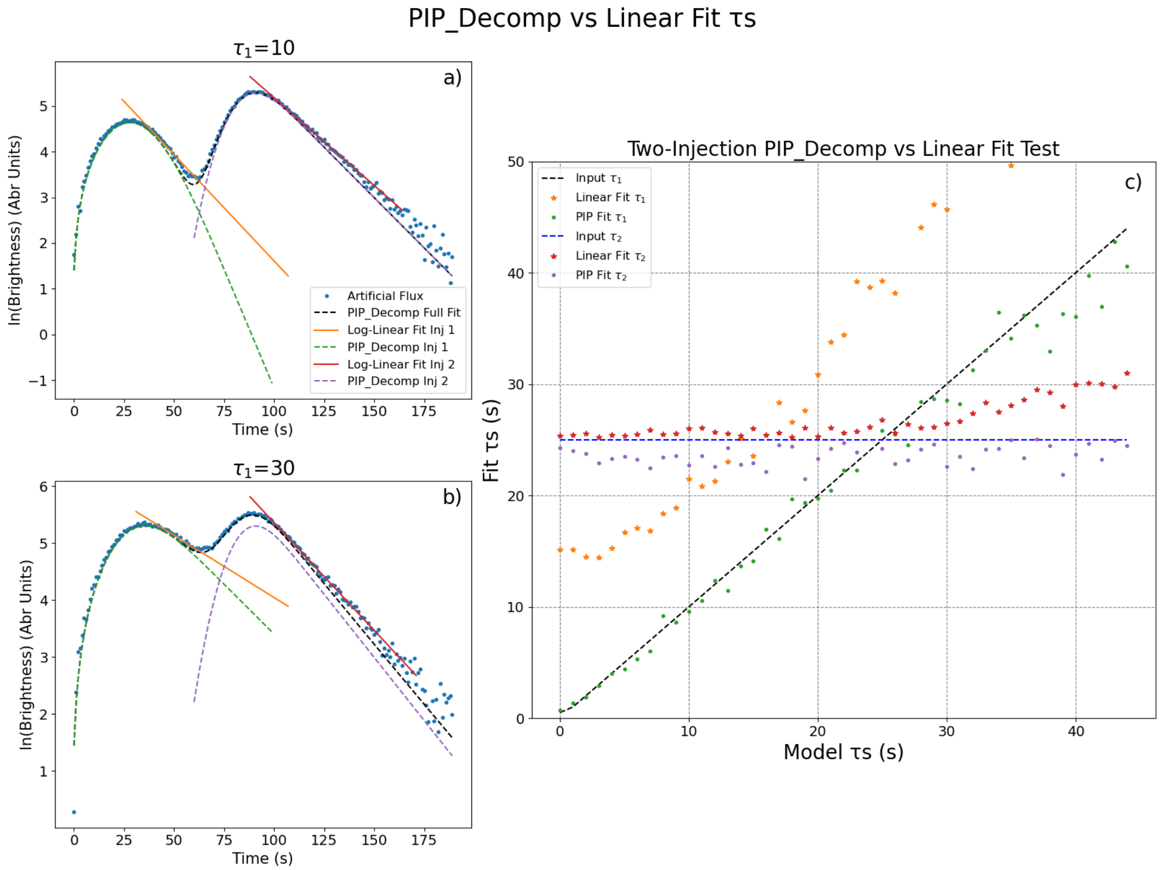

Figure 2. Results of fitting simulated 2-injection light curves, where the first injection's flux varies due to changing decay parameter τ_1. a) An example simulated light curve (points) with fits (dashed curves for each individual injection and solid curve for the sum) for τ_1=5 s, b) Same as a, for τ_1=40 s, c) The full set fitted decays τ_1 and τ_2 for PIP_Decomp (dots) and for the log-linear regression method (star symbols) as a function of varying τ_1. The lines show the decays used in the simulation.

time of the first injection becomes similar to or greater than the injection separation time, the flux of the first injection blends into that of the second (Figure 2b), making a larger apparent decay time (red points in Figure 2c) for the log-linear method. PIP_Decomp's fits of these decays do not show these errors, but rather faithfully recover the decay parameters τ_1 and τ_2 of the simulation for all values of τ_1 within the added noise.

2. COMPARING GAUSSIAN AND NON-GAUSSIAN INJECTIONS

Next we investigate the effect of PIP_Decomp's assumption of Gaussian injection shape when fitting light curves of non-Gaussian injections. This is accomplished by convolving multiple injection shapes, with similar width and integrated flux parameters, with a range of decay times and fitting the resulting light curves with PIP_Decomp.

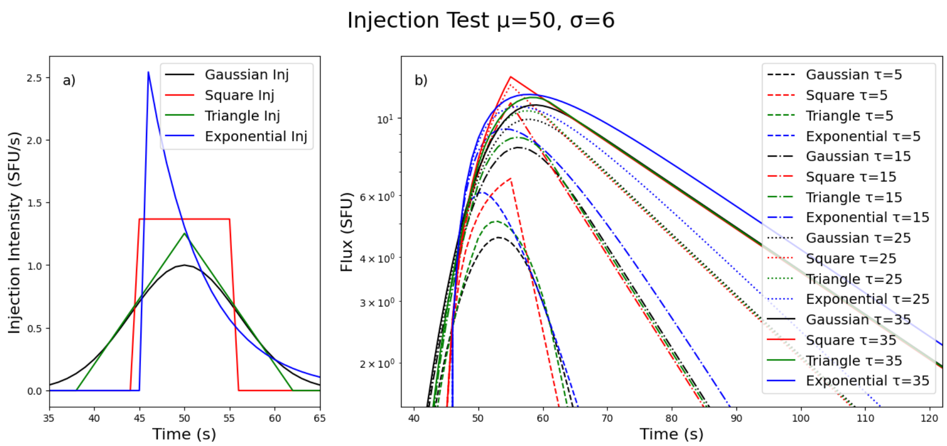

Figure 3. Non-Gaussian Injection Test - $\mu = 50$, $\sigma = 6$: a) shows a 1-pulse PIP style Gaussian injection (black) with parameters $a = 5$(SFU/s), $\mu = 50$s, $\sigma = 6$s along with the square (red), triangular (green), and exponential (blue) injections with equivalent centers of mass, normalized injection flux, and related widths to the gaussian parameters as described below. b) show the resulting flux from those injections convolved with four representative exponential decays.

This section will compare a 1-pulse PIP style injection with parameters a, μ , and σ to the light curves caused by the following injections:

- a square injection - an injection of constant amplitude for time length $2 * \sigma$ centered at $t = \mu$
- a triangular injection - a linear ramp up to a maximum amplitude over a time length of $2 * \sigma$ followed by a symmetric decay for the next $2 * \sigma$ centered at $t = \mu$

- an exponential injection - an exponential decay beginning at time $t = \mu - (ln(2) * \sigma)$ decaying at rate of $e^{(-1/\sigma)}$

Each of these non-Gaussian injections is then normalized to have a similar area as the Gaussian injection for clearer comparison (Figure 3a). Then each was convolved with a series of decays with constant $\tau = [0, 49]$s similar to the previous tests. Light curves of selected τs are shown in Figure 3b.

The key qualitative observations to take from Figure 3b are that for all four injections, and especially for the symmetric three, the light curves are very similar and remain similar with increasing decay constant τ. Some key differences that stand out:

- The square injection tends to have a sharp turnover with a brighter maximum flux than the longer term injections

- The exponential injection is front-loaded, and reaches maximum flux before the other injections. But this effect decreases with increasing τ

- The long tail of the exponential causes its brightness to turnover more slowly, and drop off more slowly than the symmetric injections

- The triangular injection and the gaussian injections have the most closely related fluxes

For more formal analysis the light curve series for each injection is then fit with PIP_Decomp in order to examine the differences between the fit parameters when attempting to fit different injections profiles (Figure 4). Each injection was fit well as a single 1-pulse PIP like flux. The square injection showed similarly small errors for a 2-pulse model, so it was fit with both.

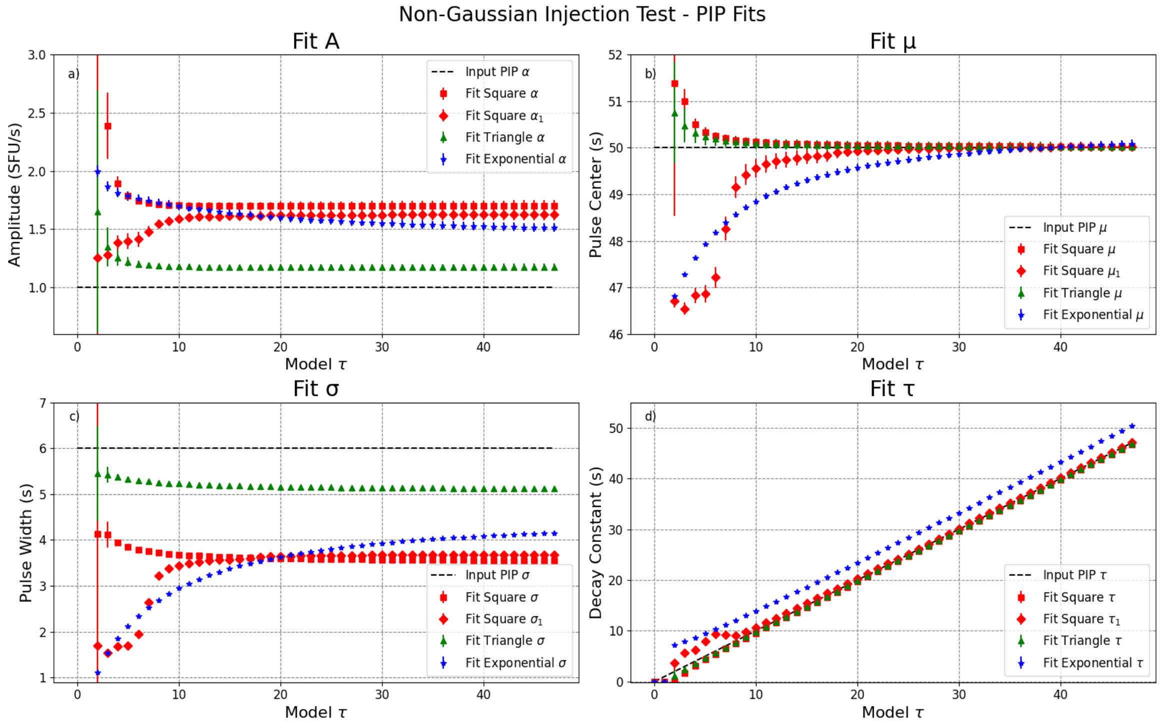

Figure 4. Non-Gaussian Injection Test- PIP Fits: shows the parameters of the previous 1-pulse PIP style injection (black), along with the PIP_Decomp Fitting parameters for the square injection as 1-pulse (red squares), for the main pulse of the square injection as 2 pulses(red diamonds), for the triangular injection (green triangle), and for the exponential (blue stars). The injections have equivalent centers of mass, normalized injection flux, and related widths to the gaussian parameters as described above: a) show fit pulse amplitudes a, b) shows fit injection centers μ, c) shows fit pulse widths σ, d) shows fit decay constants τ.

The key quantitative observations to take from the PIP_Decomp fits of shown in Figure 4 are:

- For each injection shape, μ was fit more closely with increasing τ, and tended toward the true value
- For each non-gaussian injection shape, PIP_Decomp fit the injection shape to be taller but narrower than the input gaussian. (i.e. larger a and smaller σ)
- The symmetric injections fit τ well, even when adding a second pulse to the square fit

- The exponential injection is the worst fit by the model
- The exponential injection's front loaded shape causes μ to fit earlier than expected until $\tau >> \sigma$
- The exponential injection's long tail causes τ to be over estimated by a consistent $\approx \sigma/2$ across τs

3. TESTING PIP_DECOMP: FITS OF A MODEL FLARE FROM IMAGES

With PIP_Decomp's basic capabilities demonstrated on light curves, we now turn the original goal of this work, described in the Introduction, of utilizing imaging spectroscopy data from modern radio instruments to investigate spatially varying injection and decay parameters. To test PIP_Decomp's performance for this application, an artificial flare was made of images of two injections, a spatially larger source (source 1) with a lower amplitude and relatively long impulse and decay times, and a second, more impulsive and compact source with a higher amplitude and faster-decaying temporal behavior (source 2). The presence of noise was included by adding interferometric noise of the sort produced by measurement gain errors. The level of noise was made more extreme than in typical EOVSA images to provide an easily visible level of fluctuations in the light curves for testing. A time series of images was produced for each parameter set, described below, and then the brightness in sub-sections of the images was integrated to produce two spatially distinct flux density light curves for analysis with PIP_Decomp. As shown in Figure 5a,b, the regions of interest are ROI 1 with a larger (white) box that selects isolates the spatially larger structure of source 1 alone and ROI 2, the smaller (red) box that selects the spatially confined source 2 but necessarily includes part of the source 1. As before, the use of simulated data allows for a direct check on the results in the presence of noise.

The maps were produced with two 2-D Gaussian spatial profiles with fixed spatial parameters whose amplitudes vary in time in response to their individual Gaussian

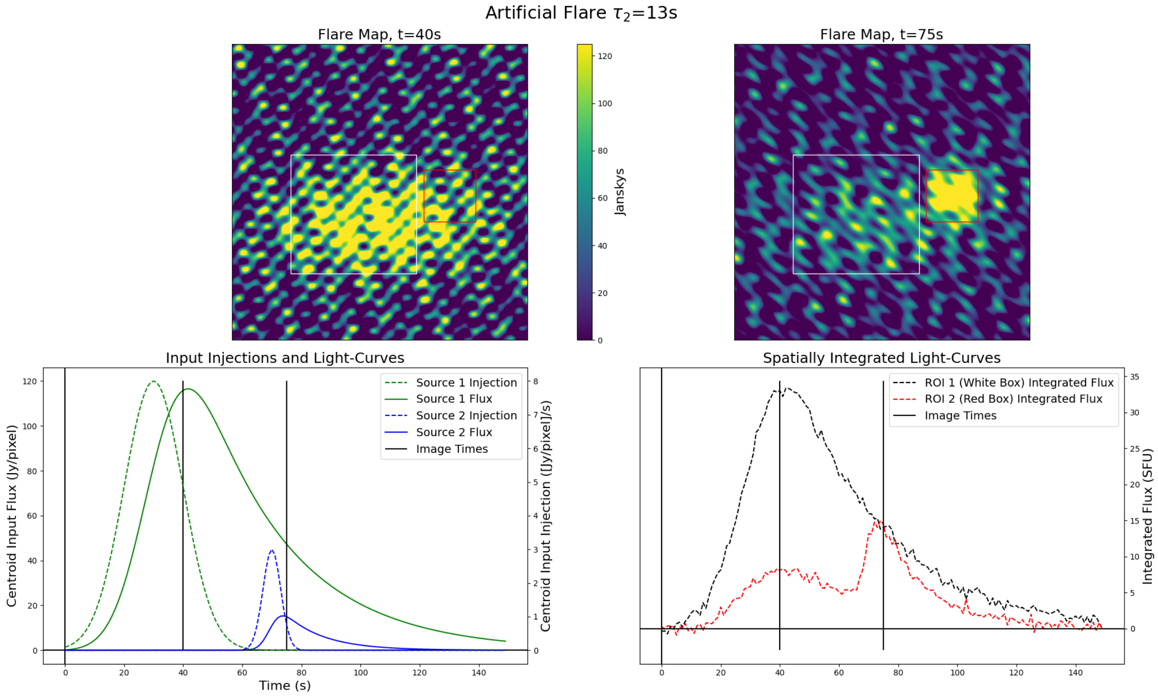

Figure 5. Simulated Flare Summary for a representative decay time ($\tau_2 = 13$ s). Two maps selected from the hundreds in the simulated flare image cube, one before (a) shows $t = 40$ s) and after (b) shows $t = 75$ s) the second injection. Two regions of interest (ROIs) are defined by the boxes in the upper panels, ROI 1 (white box) that includes the bulk of source 1 while avoiding source 2's flux, and ROI 2 (red box) that zooms in on the later source 2 but necessarily contains some of source 1's flux. The curves in c) are the ideal Gaussian injections (dashed) and resulting flux (solid) of those injections for each source at the centroid pixel of the spatial distribution. d) shows the integrated lightcurves from the ROIs at $\tau_2 = 13$ s. The vertical lines in the lower panels mark the times of the images in the upper panels. The lightcurves like those in panel d) at each τ_n will be used for fitting.

impulses and exponential decays, with time-varying spatial noise added as described above.

The artificial flare (Figure 5) is made up of these two sources placed on a 200x200 pixel grid with normalized coordinate range [-1,1], with amplitude modulated by PIP_Decomp style Gaussian impulse convolved with exponential decay. Source 1 (the larger, gradual source) is centered at $[0.2, -0.1]$ and has spatial width $\sigma_x = 0.5$, $\sigma_y = 0.25$, with injection amplitude $a_1 = 8$ SFU/s, center time $\mu_1 = 30$ s, and

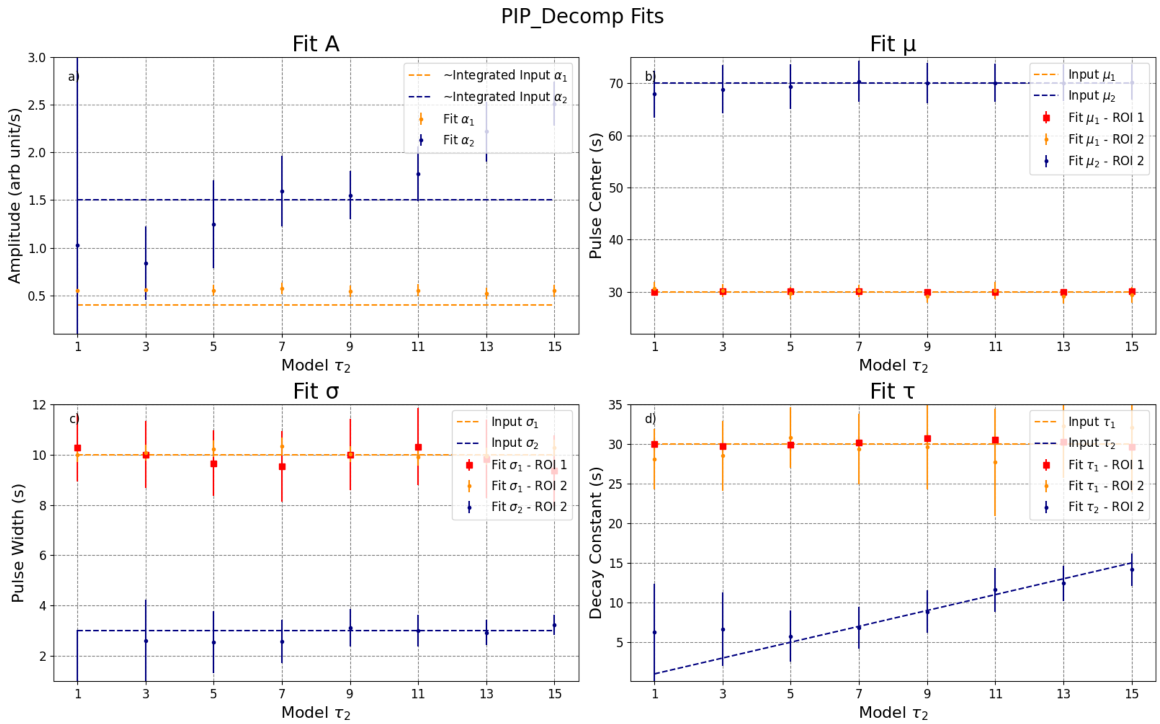

Figure 6. PIP_Decomp fits of the simulated flare, with dotted horizontal bars for the input values of the time profiles of the maps. a) shows the amplitudes (A), b) shows the injection times(μ), c) shows the injection widths (σ), d) shows the decay time constants (τ). Note that do to the spatial spreading of the full injection curve the fit amplitudes are not directly comparable to the input amplitudes.

width $\sigma_1 = 10$ s, convolved with a fixed decay $\tau_1 = 30$ s. Source 2 (the smaller, more impulsive source) is centered at $[0.0, 0.5]$ and has spatial width $\sigma_x = \sigma_y = 0.05$, with injection amplitude $a_2 = 3$ SFU/s, center time $\mu_2 = 70$s, and width $\sigma_2 = 3s$, convolved with variable decay τ_2 ranging in time from 1-15 s in steps of 2 s. Figure 5c shows the two injections (dashed lines, with amplitudes in SFU/s read from the right axis) and convolved light curves (solid lines, with amplitudes in SFU read from the left axis), for the case of $\tau_2 = 13$ s. The sources were summed with the simulated interferometric noise. The simulated interferometric noise was produced by summing a set of 12 simulated baselines at equal angles, with randomized amplitudes, such as would be produced by gain errors. The ROIs for a given time are integrated to produce a single point in each light curve, proceeding through 150 time steps to produce the entire light curves. Figure 5d shows the resulting noisy light curves after integrating over ROI 1 (dashed black curve) and ROI 2 (dashed red curve), again for the case of $\tau_2 = 13$ s, which are then fit by PIP_Decomp. The procedure is then repeated 7 times after changing the second decay time τ_2 by adding a 2 s time step at each iteration.

The example flux density light curve for $\tau_2 = 13$ s averaged over ROI 1, the region intended to contain the full event, is shown by the black dashed curve in Figure 5d. Fits of the resulting light curves for the 8 different values of delay τ_2, produced with PIP_Decomp and compared to the input data, are shown in Figure 6. The points with error bars are the parameters returned from the PIP_Decomp fits while the dashed lines are the input parameters of both sources (orange for source 1 and blue for source 2). In general PIP_Decomp was able to fit the parameters of the sources in each ROI well, that is to say for most τ_2 the fitting in general is accurate within the reported parameter errors. Although notably, the parameter errors were larger for $\tau_n \leq \sigma$, and the amplitude of source 2 was not constant against the noise